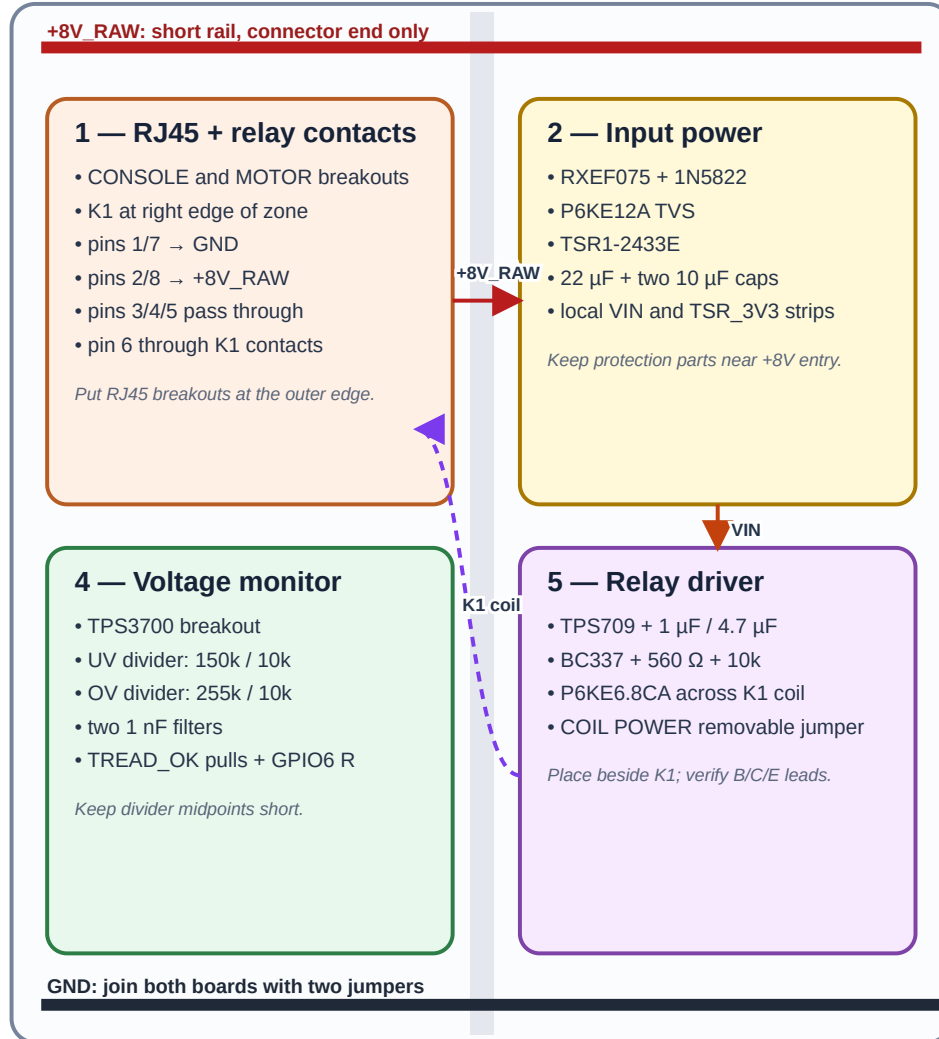


Esp32Tap Breadboard — Logical Cluster Layout

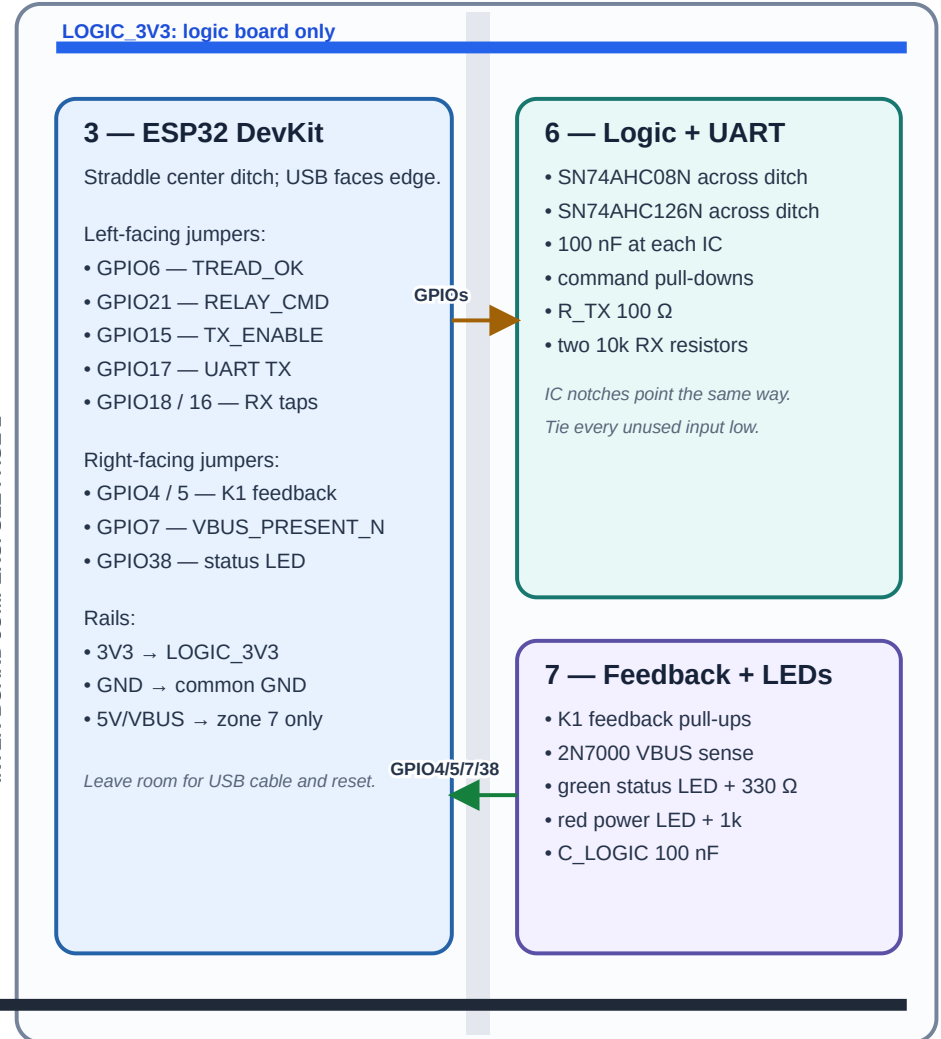
Approximate placement only. Use two full-size breadboards side-by-side; keep each zone compact, then connect zones by the named nets below. Self-contained cluster assembly cards are on pages 3–4.

Start unpowered. Leave STANDALONE POWER and COIL POWER open. Meter-identify relay, transistor, regulator, and breakout pins before inserting them.

Breadboard A — treadmill / power side



Breadboard B — ESP32 / logic side



■ BLACK GND ■ RED +8V_RAW ■ ORANGE VIN ■ BLUE 3V3 ■ VIOLET RELAY/5V ■ YELLOW CTRL/TX ■ GREEN FB/RX □ WHITE P/T

Cluster Build Order and Inter-Zone Jumpers

Build and meter-check one zone at a time. Use the point-to-point wiring PDF for every pin-level connection.

Suggested build order

Order	Zone	Stop-and-check point
1	3 — ESP32	USB boot works; 3V3 and GND rails are correct.
2	7 — Feedback/LEDs	Power LED works; GPIO pins are not shorted.
3	6 — Logic/UART	DIP orientation, VCC/GND, unused inputs, and decouplers checked.
4	1 — RJ45/K1	Pass-through continuity correct with K1 unpowered.
5	2 — Input power	Polarity checked; VIN and GND have no short. Do not join standalone power yet.
6	4 — Monitor	Divider values and TREAD_OK pull resistors meter correctly.
7	5 — Relay driver	BC337 lead order and coil TVS checked; leave COIL POWER open.

Rail policy

Rail/net	Color	Where / rule
GND	BLACK	Both boards; join with two short jumpers.
+8V_RAW	RED	Zone 1 → 2 only; do not run along logic board.
VIN	ORANGE	Zones 2, 4, 5 on power board.
LOGIC_3V3	BLUE	Zones 3, 4, 6, 7; DevKit-powered initially.
TSR_3V3	BLUE	Zone 2; removable link to LOGIC_3V3.
+5V_RLY / coil	VIOLET	Zone 5 and short K1 coil jumpers only.

Major inter-zone jumpers

Zones	Wire color	Named nets / purpose
1 → 2	RED BLACK	+8V_RAW, GND.
2 → 4	ORANGE BLACK	VIN, GND.
2 ⇄ 3	BLUE	STANDALONE POWER; install only when USB is absent.
3 ⇄ 4	BLUE YELLOW	LOGIC_3V3, TREAD_OK_MCU/GPIO6.
4 → 6	YELLOW	TREAD_OK hardware permission.
3 → 6	YELLOW	RELAY_CMD, TX_ENABLE, UART TX.
6 → 5	YELLOW	RELAY_GATE.
5 ⇄ 1	VIOLET	K1 coil pins 1/16; keep short.
6 → 1	YELLOW	R_TX output to K1 pin 8.
1 → 3	GREEN	CONSOLE pin 6 → GPIO18; pin 3 → GPIO16.
1 → 7	GREEN	K1 pins 9/11/13 feedback.
7 ⇄ 3	GREEN VIOLET BLUE	GPIO feedback/sense, 5V/VBUS, 3V3.

Fast physical rules

- Put every DIP IC across a breadboard center ditch, with notches pointing the same direction.
- Put each 100 nF capacitor directly beside its IC supply pins.
- Keep UV_SENSE and OV_SENSE resistor/capacitor nodes inside zone 4.
- Keep BC337, its two resistors, coil TVS, and K1 coil pins together.
- Reserve white wires for RJ45 pins 3/4/5 direct pass-through.
- Label both ends of every jumper that crosses between boards.

This page groups the circuit; it does not replace the 126-row From/To wiring list. Never infer a physical pin from package shape—meter or datasheet-verify it.

Cluster Assembly Cards — Zones 1–4

All order references are from DigiKey invoice **129832848**. Use one card while assembling each cluster. **NOT ORDERED** means absent from that invoice.

1 — RJ45 breakouts + relay contacts

Part	Qty / description	DigiKey part number
K1	1 × G5V-2 DC5 DPDT relay	39-G5V-2DC5-ND
RJ45 breakouts	2 × CONSOLE/MOTOR	NOT ORDERED

Assemble here: RJ45 pins 1/7 to GND, 2/8 to +8V_RAW, pins 3/4/5 straight through; pin 6 uses K1 pins 4/6/8. Put K1 at the zone 1/5 boundary.

Wire: BLACK GND RED +8V WHITE PASS-THROUGH YELLOW PIN 6/TX GREEN FEEDBACK

2 — Protected input power

Part	Qty / description	DigiKey part number
F1	1 × RXEF075 resettable fuse	RXEF075HF-ND
D1	1 × 1N5822-TP Schottky	1N5822-TPMSCT-ND
TVS	1 × P6KE12A-TP	P6KE12A-TPMSCT-ND
DC/DC	1 × TSR 1-2433E	1951-TSR1-2433E-ND
Bulk cap	1 × 22 µF 25 V aluminum electrolytic, polarized	732-8821-1-ND
TSR caps	2 × 10 µF 50 V aluminum electrolytic, polarized	1189-2322-ND
Power jumper	1 × removable STANDALONE POWER	NOT ORDERED

Assemble here: +8V_RAW → RXEF075 → 1N5822 → VIN. P6KE12A from VIN to GND. Electrolytic body stripe is negative.

Wire: RED +8V_RAW ORANGE VIN BLUE TSR_3V3 BLACK GND

3 — ESP32 DevKit

Part	Qty / description	DigiKey part number
DevKit	1 × ESP32-S3-DEVKITC-1-N8R8	1965-ESP32-S3-DEVKITC-1-N8R8-ND

Assemble here: straddle the center ditch with USB facing the board edge. DevKit 3V3 feeds LOGIC_3V3 during USB bring-up; DevKit GND joins common GND.

Wire: BLUE 3V3 BLACK GND YELLOW OUTPUT/CONTROL GREEN INPUT/FEEDBACK VIOLET 5V/VBUS

4 — TPS3700 voltage monitor

Part	Qty / description	DigiKey part number
Monitor	1 × TPS3700DDCR	296-30395-1-ND
Adapter	1 × LCQT-SOT23-6	A879AR-ND
Header	Cut from PRPC040SAAN-RC	S1011EC-40-ND
UV top	1 × 150 kΩ 1% axial	150KXBK-ND
OV top	1 × 255 kΩ 1% axial	13-MFR-25FBF52-255K-ND
10 kΩ	3 × UV/OV bottoms + TREAD_OK pull-up	NOT ORDERED
100 kΩ	1 × TREAD_OK pull-down	NOT ORDERED
4.7 kΩ	1 × GPIO6 isolation	NOT ORDERED
Filter caps	2 × 1 nF 50 V C0G ceramic radial, non-polar	445-173473-1-ND
Bypass cap	1 × 100 nF 50 V X7R ceramic radial, non-polar	BC1084CT-ND

Assemble here: keep UV_SENSE and OV_SENSE parts close to the adapter; tie TPS3700 outputs together at TREAD_OK.

Wire: ORANGE VIN BLUE 3V3 BLACK GND YELLOW TREAD_OK WHITE LOCAL SENSE

Cluster Assembly Cards — Zones 5–7

All capacitor construction, polarity, and purchase data is inline. Use equal or higher voltage ratings only.

5 — Relay supply and driver

Part	Qty / description	DigiKey part number
Regulator	1 × TPS70950DBVR	296-35484-1-ND
Adapter	1 × LCQT-SOT23-6	A879AR-ND
Header	Cut from PRPC040SAAN-RC	S1011EC-40-ND
Q relay	1 × BC337-40	4878-BC337-40CT-ND
Base resistor	1 × 560 Ω 1% 1/4 W axial	MFR-25F8F52-560R-ND
Base pull-down	1 × 10 kΩ	NOT ORDERED
Coil TVS	1 × P6KE6.8CA bidirectional	18-P6KE6.8CACT-ND
Input cap	1 × 1 μF 25 V X7R ceramic radial, non-polar	445-173583-1-ND
Output cap	1 × 4.7 μF 50 V aluminum electrolytic, polarized	732-8660-1-ND
Coil jumper	1 × removable COIL POWER	NOT ORDERED

Assemble here: verify BC337 B/C/E leads; put TVS directly across K1 pins 1/16. Electrolytic body stripe is negative. Leave COIL POWER open.

Wire: ORANGE VIN VIOLET +5V/COIL YELLOW RELAY_GATE BLACK GND

7 — Relay feedback, VBUS sense, and LEDs

Part	Qty / description	DigiKey part number
VBUS MOSFET	1 × 2N7000	4878-2N7000CT-ND
Green LED	1 × LTL-4233	160-1130-ND
Green resistor	1 × 330 Ω	NOT ORDERED
Red LED	1 × 151051RS11000	732-5016-ND
Red resistor	1 × 1 kΩ	NOT ORDERED
10 kΩ	4 × feedback/VBUS pulls	NOT ORDERED
Logic bypass	1 × 100 nF 50 V X7R ceramic radial, non-polar	BC1084CT-ND

Assemble here: meter-identify 2N7000 G/D/S. LED long lead is anode; short/flat-side lead goes toward GND.

Wire: BLUE 3V3 VIOLET 5V/VBUS GREEN FEEDBACK/SENSE YELLOW LED SIGNAL BLACK GND

6 — Hardware gating and UART

Part	Qty / description	DigiKey part number
AND gates	1 × SN74AHC08N PDIP-14	296-4524-5-ND
Buffer	1 × SN74AHC126N PDIP-14	296-4535-5-ND
Bypass caps	2 × 100 nF 50 V X7R ceramic radial, non-polar	BC1084CT-ND
10 kΩ	4 × command pulls + two RX taps	NOT ORDERED
TX resistor	1 × 100 Ω series	NOT ORDERED

Assemble here: both ICs straddle the ditch with notches aligned. Put each 100 nF capacitor at pins 14/7. Tie every unused input low.

Wire: BLUE VCC BLACK GND YELLOW CONTROL/TX GREEN RX

Common bench supplies

Item	Quantity / use	DigiKey order status
Full-size breadboards	2, side-by-side	NOT ORDERED
Jumper-wire assortment	For 126 From/To connections	NOT ORDERED

Color convention: BLACK GND RED +8V ORANGE VIN BLUE 3V3 VIOLET RELAY/5V YELLOW CONTROL/TX GREEN FEEDBACK/RX WHITE PASS-THROUGH

Label both ends of every inter-board jumper. If a suggested color is unavailable, use another color and label both ends. Wire color is identification, not an electrical rating.

Start unpowered. Leave STANDALONE POWER and COIL POWER removed until their bring-up steps explicitly install them.